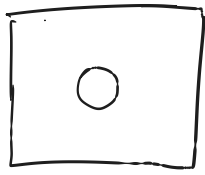


Assignment2

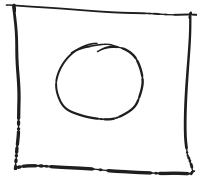
Theoretical Workout

In Problem 04 of Assignment_01, find out the number of pixels per square inch (DPI) for both the images taken at 50 cm and 80 cm respectively.

Assignment -2



80cm away



50 cm away

dim of both = 960×1280

Goal is to find the no. of pixels per sq. inch

How many pixels in this image represents

Abstract

1. finding dpi
2. Reading number of lines
3. Diff thresholding algorithm implement
4. Filter apply, thresholding, draw random circles
5. Equalization implementation

Concepts

Q1. Find the number of pixels per square inch (DPI) for both images taken 50cm and 80cm.



Spatial resolution $\Rightarrow$ DPI how many pixels in the photo represent 1 inch in real-world circle. Essentially meaning how much detail of the object is being captured per unit.

1. Find radius of circles in pixels (r)
2. convert radius of actual circle = 5cm $\rightarrow$ 1.9685 in (R)

DPI of an image = Radius in pixels / Radius in inches

$\rightarrow$ Area of circles in pixels:

$$A_{50} = \pi r_{50}^2 = 32297$$

$$r_{50} = \sqrt{\frac{32297}{\pi}} = 101.39$$

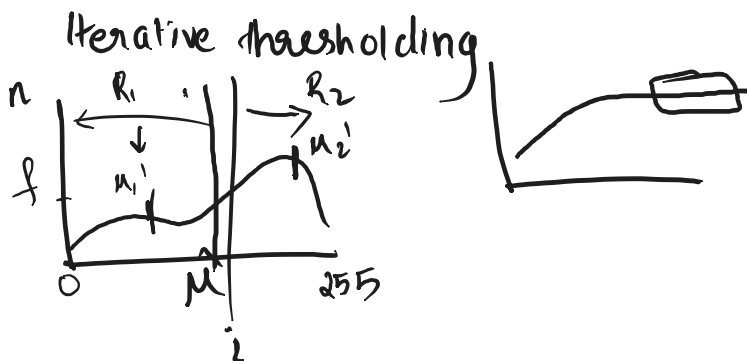
$$\therefore \text{DPI}_{50} = \left(\frac{101.39}{1.9685} \right)^2 = (51.51)^2 = \underline{\underline{2653.28 \text{ px/in}^2}}$$

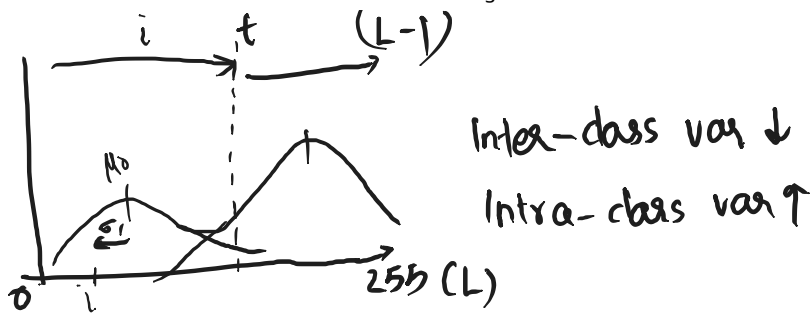
$$A_{80} = \pi r_{80}^2 = 12206$$

$$r_{80} = \sqrt{\frac{12206}{\pi}} = 62.33$$

$$\therefore \text{DPI}_{80} = \left(\frac{62.33}{1.9685} \right)^2 = (31.66)^2 = \underline{\underline{1002.6 \text{ px/in}^2}}$$

Q3. Use different thresholding algorithms on the following image to generate and save the binary image after thresholding.





$H \rightarrow$ normalized hist

$L \rightarrow$ total bins

$C_0 \rightarrow$ BG pixels

$0 \rightarrow t-1$

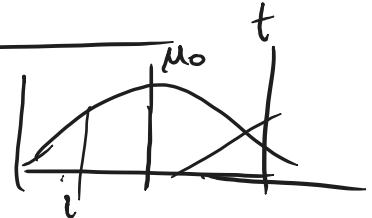
$C_1 \rightarrow$ FG pixels

$t \rightarrow L-1$

$C_0 \rightarrow P(i|C_0)$

$C_1 \rightarrow P(i|C_1)$

$$(i - \mu_0)^2$$



$$\sigma_0^2(t) = \sum_{i=0}^{t-1} (i - \mu_0)^2 P(i|C_0) / \left(\frac{f(0 \rightarrow t)}{\text{total no. px}} \right)$$

$\sigma_T^2(t) \rightarrow$ variance of the hist

$\mu_T \rightarrow$ overall mean

$\mu_0 \rightarrow$



$$\sigma_0, \sigma_1 \uparrow$$

$$\sigma_b \downarrow$$

Otsu's thresholding

So basically Otsu's thresholding is a clustering based method. Assumes the image contains foreground and background pixels of the image.

Calculating optimal threshold that
separate the two ->
by increasing intra-class variance and
reducing inter-class variance.

Otsu's thresholding

Aim: To find the best t that separates foreground & background as if its bimodal.

Math

$$L = [0, 255]$$

$$i = \text{Pixel}$$

N = Total no. of pixels

n_i = no. of pixels with intensity i

$$P_i = \frac{n_i}{N} = \text{Prob. of intensity } i$$

Class probabilities (wts)

$$w_0(t) = \sum_{i=0}^t P_i \quad w_1(t) = \sum_{i=t+1}^{L-1} P_i = 1 - w_0(t)$$

Class means:

$$\mu_0(t) = \sum_{i=0}^t \frac{i \cdot P_i}{w_0(t)} \quad \& \quad \mu_1(t) = \sum_{i=t+1}^{L-1} \frac{i \cdot P_i}{w_1(t)}$$

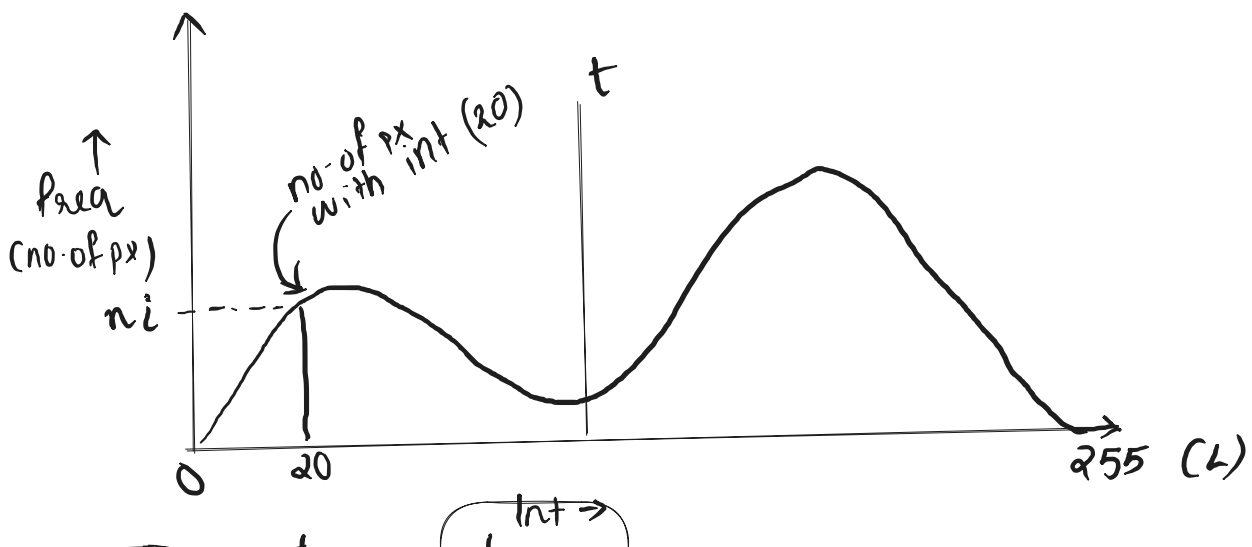
Total mean intensity:

$$\mu_T = \sum_{i=0}^{L-1} i \cdot P_i$$

$$\mu_T = w_0(t) \mu_0(t) + w_1(t) \mu_1(t)$$

Between-class var. $\rightarrow$ maximize

$$\sigma_b^2(t) = w_0(t) w_1(t) [\mu_0(t) - \mu_1(t)]^2$$



$$w_0(t) = \sum_{i=0}^t p_i = \left(\sum_{i=0}^t \frac{n_i}{N} \right) \quad (1)$$

$$w_1(t) = \sum_{i=t+1}^{L-1} p_i = \sum_{i=t+1}^{L-1} \left(\frac{n_i}{N} \right)$$

$$\mu_0(t) = \frac{\sum_{i=0}^t i \cdot p_i}{w_0(t)}$$

$$\mu_1(t) = \frac{\sum_{i=t+1}^{L-1} i \cdot p_i}{w_1(t)}$$

$$\left(\frac{\sum_{i=0}^t i \cdot \frac{n_i}{N}}{\sum_{i=0}^t \frac{n_i}{N}} \right) \quad (2)$$

Total mean intensity:

$$\mu_T = \sum_{i=0}^{L-1} i \cdot p_i = w_0(t) \mu_0(t) + w_1(t) \mu_1(t)$$

Between-class variance (maximize this)

$$\sigma_b^2(t) = \underbrace{w_0(t)}_{(1)} \underbrace{w_1(t)}_{(2)} [\mu_0(t) - \mu_1(t)]^2$$

Question 4

Draw a fence of an arbitrary shape by hand (!) on a clean paper with white background and printed lines on it as explained in the class.

Take a photo of this image.

Write a program:

- to read the photographed image and translate it to a gray scale image of size say, 300 x 300 of this image.
- Use global thresholding algorithm to clean image to remove all unwanted lines on it if any.

- c) Save original and binary image by name fence_original.jpg and fence_threshold.jpg.
- d) Write a program to draw a filled circle of radius 5 randomly in this image (fence_threshold.jpg). Generate 50 such images. Store these images into two folders as explained in the class.

Q4) Fence

Filter logic input = 1280×1280 out = 300×300

$$\text{Ratio} = \frac{\text{og_size}}{\text{New size}} = \frac{1280}{300} = 4.267$$



7x7 with 3x3 mean filter $\rightarrow$ 5x5 output

$$\left(\frac{7-3}{\text{stride}} \right) + 1 = \frac{4}{2} + 1 = 3 //$$

$$\therefore \left(\frac{n_{\text{row}} - f_{\text{row}}}{\text{stride}} \right) + 1 = \text{output}$$

$$300 = \left(\frac{1280 - f_{\text{row}}}{\text{stride}} \right) + 1 \Rightarrow 299 = \frac{1280 - f_{\text{row}}}{\text{stride}}$$

Prefer less padding $\Rightarrow$ stride $\sim$ frow

$$299 \times \text{stride} = 1280 - f_{\text{row}}$$

If stride = 4

$$1196 = 1280 - f_{\text{row}}$$

f $\rightarrow$ any size
stride $\rightarrow$ Integer

$$\text{So filter len} = 1280 - 1196$$
$$f_{\text{col}} = f_{\text{row}} = \underline{\underline{84}}$$
